EV CHARGING · NETWORK INTRUSION DETECTION

EVNet Sentinel

A leakage-corrected reproduction and comparative study of intrusion detection on EV charging networks — with a closed-loop response

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Watch the video

Code & results

99.8%
accuracy from **one timestamp column** alone — the published 98.4% rests on a leak

0.558
honest **macro-F1** with the leak removed (± 0.008, 3 seeds)

≥ 99.8%
floods named exactly (95% lower bound) — scans named only 34%

20 / 20
recorded attacks **contained** in the response replay; 0 actions on normal traffic

1 Problem

Every public EV charger (EVSE) is a networked computer. It reports to a **charging management system (CSMS)** over OCPP, usually across the site's shared wifi, and talks to the car over ISO 15118. Flooding a charger cuts its link to the CSMS: sessions cannot be authorised or billed.

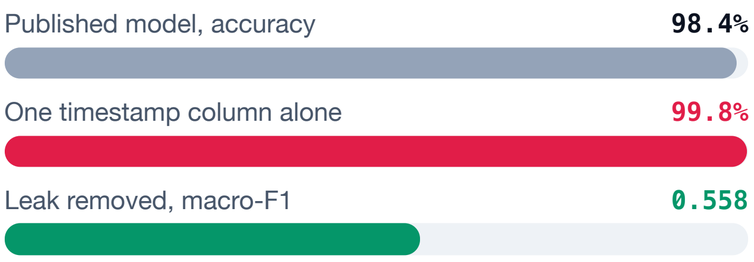
Makhmudov et al. [1] reported **98.4% accuracy** with an adaptive random forest and drift detection. We set out to reproduce it, and to go beyond detection.

2 Dataset

CICEVSE2024 (Canadian Institute for Cybersecurity): traffic from two real charging stations, idle and charging, recorded from a mirrored switch port.

- **1,198,151** flows after de-duplication, **15** classes, **60** flow features
- Attacks: 5 volumetric floods, 6 nmap scan types, ICMP and Slowloris
- Senders: a laptop and a Raspberry Pi on the wifi, and a **compromised car** scanning over the charging cable

3 The leak



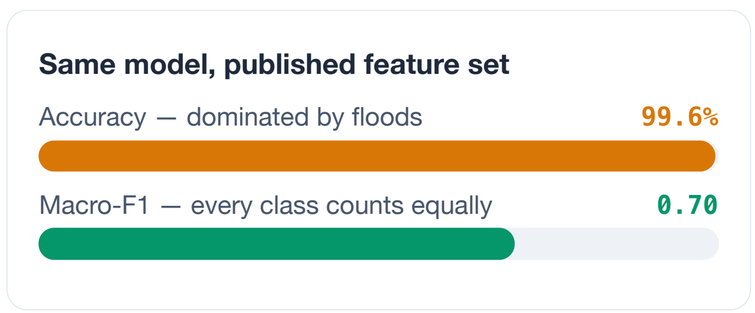
> Six absolute capture timestamps identify the **recording**, not the traffic: a single column (`bidirectional_first_seen_ms`) out-scores the published model. The drift alarms also land on capture-file seams, not real concept drift.

4 Method

1. **Remove leakage** — drop the timestamps and `src_port`; de-duplicate flows.
2. **Split** — stratified train / validation / test, repeated for seeds 42, 1337, 2024.
3. **Detect** — six static models (XGBoost, LightGBM, Random Forest, Decision Tree, Logistic Regression, SVM) and an online ARF + ADWIN.
4. **Test per attack** — every class, every model, 30 × 50 unseen flows per model set; graded detected / right category / named.
5. **Respond** — a containment policy, replayed over every recorded attack (section 8).

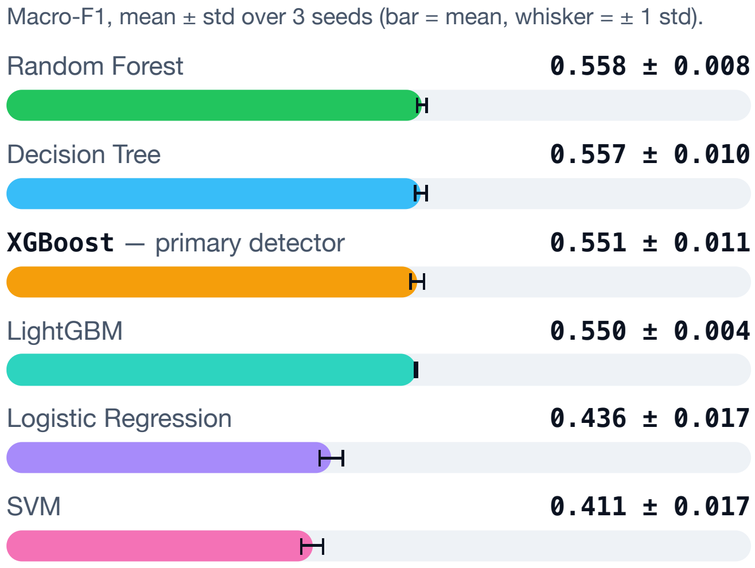
5 Why macro-F1, not accuracy

After de-duplication, floods are **over 80%** of the flows (99% on the published feature set). A model that names every flood and misses every scan still scores high accuracy.



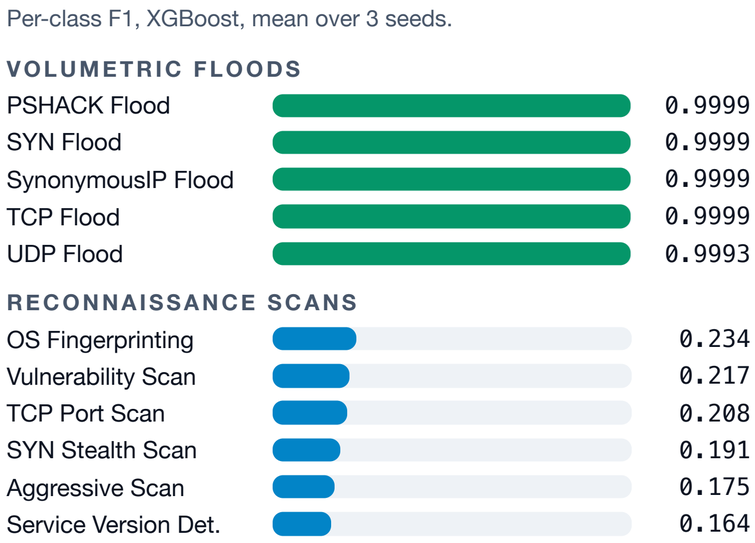
Macro-F1 averages the F1 of each of the 15 classes, so a missed scan type weighs as much as a flood.

6 Corrected results



The four tree models tie within one standard deviation. **XGBoost** is the primary detector: the fewest false alarms on normal traffic, **1.8%** (Random Forest 2.6%).

7 Floods are solved. Scans are not.



The six nmap modes look almost identical at flow level. Still, **96%** of scans are recognised *as scans* — the response acts on that.

8 Closing the loop

Detection alone does not stop an attacker. Sentinel recommends an action matched to how reliable the detector is; **a person approves every action**, and the chargers and CSMS can never be cut off.

Detect	Contain	Harden	Verify
classify each flow	one action per device	fix it for next time	attack flows stop

ATTACK	ACTION	WHO · HOW
Flood from the wifi	Quarantine	Analyst · switch port / ACL
Scan from the wifi	Restrict	Analyst · no access to chargers or CSMS; device watched
Scan from a compromised car	Stop session	CSMS operator · RemoteStopTransaction, block the vehicle's ID token
Watched device starts flooding	Quarantine	Escalation at a much lower bar

9 Response, replayed

Every recorded attack, from the device that really sent it, streamed through the policy with XGBoost's real verdicts and an operator who approves after 5 flows.

20/20
attacker–attack pairs contained

11 → 1
flood flows to quarantine, cold vs after a scan

0
actions over 2,000 benign flows

A replay of recorded traffic through the policy, not live blocking: Sentinel is a passive, mirror-port IDS. The dataset has only 12 distinct benign flows, so the zero is a small sample.

10 Conclusions

- The published 98.4% comes from **capture timestamps**; corrected macro-F1 is **≈ 0.55** for every tree model.
- **Floods are solved** (≥ 99.8% named); **scans are data-limited**, not model-limited — stronger models do not lift the ceiling.
- A detector earns its keep in the **response**: category-level actions contain every recorded attack.

> **Next:** session-level features to separate the scan types, and validation on a live OCPP testbed.

11 References

[1] F. Makhmudov, D. Kilichev, U. Giyosov, F. Akhmedov, "Online machine learning for intrusion detection in electric vehicle charging systems," *Mathematics*, 13(5):712, 2025.
[2] Canadian Institute for Cybersecurity, "CICEVSE2024 dataset," University of New Brunswick, 2024.
[3] J. Montiel et al., "River: Machine learning for streaming data in Python," *JMLR* 22(110), 2021.

Team

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